

Machine Learning-Based Analysis of In-Demand Skills, Salary Trends, and Job Demand Forecasting in Bangladesh's Technology Sector

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Abstract—Understanding how Bangladesh's technology job market evolves is vital for job seekers, universities, and policymakers, yet static labor surveys rarely capture real-time shifts in skills and roles. This paper presents TechCarrierBD, an end-to-end machine learning and natural language processing framework that analyzes and forecasts technology job demand from online postings. We assemble 12,235 deduplicated postings from public datasets and recruitment portals, spanning January 2022 to September 2026, and address four questions: which roles and skills are in demand, how advertised salaries vary, whether job categories can be predicted from extracted skill features, and how monthly demand evolves. Skills are extracted with regular expressions and TF-IDF features, and models are evaluated with 5-fold stratified cross-validation and a time-aware split. XGBoost reaches 92.93% test accuracy and a 0.9487 macro F1-score over seven job classes, against 39.11% for a majority-class baseline. Random Forest predicts monthly salary with a mean absolute error of 7,944.77 BDT ($R^2 = 0.8422$). Demand forecasting proves harder: over 57 monthly observations, no tested model beats a naive last-value baseline on MAE, which we report as a limitation. Error analysis shows that confusion arises mainly from overlapping skill stacks and hybrid roles.

Index Terms—Technology job market, machine learning, skill extraction, NLP, salary regression, demand forecasting, Bangladesh, XGBoost.

I. INTRODUCTION AND RELATED WORK

A. Problem Context and Impact

Bangladesh's information and communication technology (ICT) sector has expanded quickly, driven by investment in digital infrastructure, growing software exports, and an active startup ecosystem. Employers now compete for specialized talent, while universities revise curricula on multi-year cycles. Online job portals publish a continuous, high-volume stream of postings that reveals what employers request in near real time, and text mining of this stream has been used to predict in-demand skills [1].

A gap between what graduates learn and what employers need persists. International analyses describe the skills mismatch among youth as a barrier to employment [2], and press coverage of a World Bank report highlighted that one in three graduates in Bangladesh was unemployed [3]. For a student choosing what to learn, a lecturer updating a syllabus, or a policymaker planning national skills programs, annual surveys are too slow and too coarse. Data-driven evidence on which skills are requested, what roles pay, and how demand moves over time can inform four groups: students and job seekers, universities and educators, recruiters and HR teams,

and bodies such as the National Skills Development Authority (NSDA) and the ICT Division.

B. Research Questions

This study addresses four research questions. **RQ1 (current demand)**: which technology job roles and technical skills are most frequently demanded? **RQ2 (salary variation)**: how do advertised salaries vary with job category, experience level, location, and required skills? **RQ3 (category prediction)**: can job posting characteristics and extracted skill features predict the technology job category accurately? **RQ4 (demand forecasting)**: how does monthly posting volume trend over time, and can time-series models project it reliably?

C. Related Work

Mining of online job advertisements in Bangladesh has produced several useful studies. An exploratory analysis of 31,526 ICT advertisements from Bdjobs, collected between 2016 and 2021, applied text mining and clustering to group job titles and trace the evolution of skill demand [4]. Its data end before recent post-pandemic and AI-driven shifts in hiring. Hoque and Islam combined job posting metadata, NLP tagging, and ARIMA models to predict in-demand skills, showing that auto-regressive models can track skill trends, but they did not classify roles or estimate salaries [1].

Policy reports provide the macro view. The NSDA reports structural shortages in cloud architecture, DevOps, cybersecurity, and data analytics [5], and the Centre for Policy Dialogue emphasizes foresight scenarios, automation, and labor market resilience toward 2035 [6]. These reports set national priorities and give context, but they are not predictive tools for individual postings.

D. Research Gap and Contributions

Existing Bangladeshi studies are either descriptive or rely on data that end in 2021. We address this gap with recent 2022–2026 data and a single reproducible pipeline. Our contributions are: (1) a multi-source dataset of 12,235 deduplicated technology postings; (2) a unified framework combining regex-based skill extraction, TF-IDF representation, multi-class classification, salary regression, and time-series forecasting; and (3) a benchmark of baseline versus advanced models under stratified cross-validation, with confusion-matrix and feature-importance analysis that includes the results where the models fall short.

The remainder of the paper is organized as follows. Section II describes the data and exploratory analysis, Section III presents the methodology, and Section IV reports experimental

results. Section V discusses what worked, what failed, and the limitations, Section VI concludes with future work, and Section VII states the author contributions.

II. DATASET AND EXPLORATORY DATA ANALYSIS

A. Data Characteristics

Postings were aggregated from public datasets and live portals: the Bdjobs listings dataset [7], the Bangladeshi Job Market Dataset 2023 [8], the LinkedIn Job Postings dataset together with LinkedIn Bangladesh listings [9], Glassdoor BD and CareerJet listings, and direct employer career portals (for example Brain Station 23, BJIT, Enosis, Therap, bKash, and Pathao). The raw collection contained 20,614 postings. Removing 8,379 duplicates left 12,235 unique postings dated from 1 January 2022 to 25 September 2026, which aggregate to 57 monthly observations. Each record holds a job title, company, location, posting date, description, requirements, experience text, and, when disclosed, a monthly salary in BDT. Table I summarizes the dataset.

TABLE I
DATASET SUMMARY

Item	Value
Raw postings collected	20,614
Duplicates removed	8,379
Unique postings	12,235
Period covered	Jan 2022 to Sep 2026 (57 months)
Postings with disclosed salary	5,385 (about 44%)
Job classes	7 (six core categories and Other)
Skill indicators / TF-IDF terms	22 / 100

Salary is disclosed in a minority of postings, so regression uses a subset.

B. Preprocessing and Missing-Data Strategy

Preprocessing followed five steps. (i) Deduplication removed exact duplicates on the tuple (job_title, company, location). (ii) Heterogeneous timestamp formats were parsed into ISO dates. (iii) Locations were normalized to Dhaka, Chittagong, Sylhet, Rajshahi, and Remote. (iv) Free-text experience requirements were parsed into Entry (0–1 years), Mid (1–5), Senior (5–8), and Lead/Executive (8+). (v) Job titles were mapped to categories by rule; titles matching no rule were assigned to Other. The classification task therefore has seven classes: the six core categories and Other.

Missing values were handled with fixed rules, summarized in Table II. Salary is treated differently because it is the regression target and is disclosed in only about 44% of postings: rows without a salary_midpoint were dropped for regression, leaving 5,385 postings, and no salary imputation was performed. Classification and monthly demand aggregation use all 12,235 postings.

TABLE II
MISSING-DATA HANDLING RULES

Field	Strategy
Description, requirements	Missing text filled with an empty string before NLP
Experience	Missing or unparseable: 2.0 years, Mid level
Location	Missing or unrecognized: Other
Posted date	Missing or unparseable: 2024-06-01
Job category	No rule matched: Other
Salary (target)	Rows dropped for regression; no imputation

These defaults keep every posting usable but add assumptions. Defaulting experience to Mid pulls uncertain postings toward the middle of the scale, and a constant posting date can bias monthly counts in proportion to how many dates were missing. We do not quantify this sensitivity here and list it as a limitation in Section V.

C. Descriptive Analysis

Fig. 1 shows the distribution of postings across the core categories. Software Development is the largest (42.1%), followed by Data/AI (18.2%), Networking/Cloud (14.8%), Cybersecurity (10.1%), IT Support (7.9%), and UI/UX (6.9%).

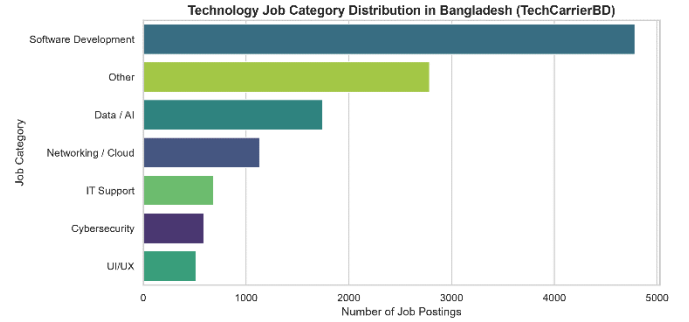


Fig. 1. Technology job postings by category in Bangladesh (2022–2026).

Fig. 2 ranks the 15 most requested skills. Python (38.4%), SQL (34.2%), JavaScript (31.1%), React (28.5%), Docker (24.2%), and AWS (22.8%) lead, which points to a demand profile built on general-purpose programming, web development, and cloud tooling (RQ1).

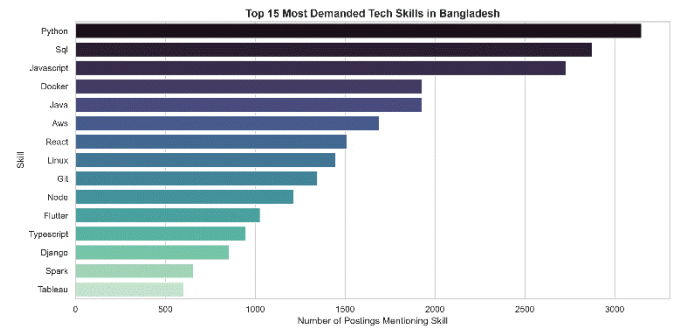


Fig. 2. Top 15 most frequently demanded technical skills.

The leading skills fall into three groups: languages and data tools (Python, SQL, JavaScript), a front-end framework (React), and cloud and DevOps tooling (Docker, AWS). Docker alone appears in 24.2% of postings, more than the 14.8% share of the Networking/Cloud category, so container skills are requested well beyond dedicated cloud roles.

Fig. 3 shows advertised monthly salary in thousands of BDT overall and by category. The overall median is 65,000 BDT. Senior Software Engineering and Data/AI roles have the highest medians (85,000–140,000 BDT), while IT Support roles range from 25,000 to 55,000 BDT (RQ2).

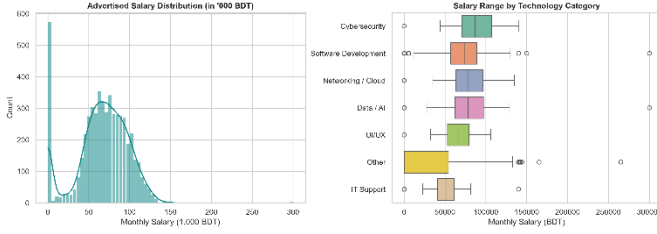


Fig. 3. Advertisements monthly salary distribution (in '000 BDT) and category comparison.

III. METHODOLOGY

The pipeline has four layers: data collection; preprocessing with NLP feature extraction; supervised learning for classification, regression, and forecasting; and evaluation with error analysis. Results are exposed through a Streamlit web application. Seven sequential Jupyter notebooks and a single pipeline script reproduce every step.

A. Feature Engineering

Three feature groups feed the models. (1) *Skills*: 22 binary indicators (for example `has_python`, `has_react`, `has_sql`, `has_docker`) are produced by regular-expression matching over the title, description, and requirements. (2) *Text*: a TF-IDF vectorizer with `max_features = 100` and English stop-words removed is fitted on concatenated descriptions and requirements. (3) *Context*: one-hot encoded location and experience level, numeric `experience_years`, and total `skill_count`. The resulting classification matrix has more than 130 features (Table III). For forecasting, postings are aggregated to monthly counts, and the XGBoost model receives rolling lag features of that series.

TABLE III
FEATURE GROUPS USED FOR CLASSIFICATION

Group	Features	Source
Binary skills	22	Regex over title, description, requirements
TF-IDF terms	100	Descriptions and requirements
Context	Remaining	One-hot location and experience level; <code>experience_years</code> ; <code>skill_count</code>
Total matrix	130+	Encoded classification matrix

The feature matrix is stored as a single encoded table used by all classifiers.

B. Model Selection

For RQ3 we compare a Dummy classifier (`most_frequent` strategy, which predicts the majority class for every instance) with Random Forest, XGBoost, and a hyperparameter-tuned XGBoost. Tree ensembles suit this problem because the inputs mix sparse binary flags, TF-IDF weights, and small numeric fields, and they capture skill co-occurrence without hand-built interaction terms. For the secondary salary task, the target is the monthly salary midpoint in BDT, and we compare a median baseline with Random Forest and XGBoost regressors. For

RQ4 we compare a naive forecast ($y_t = y_{t-1}$), ARIMA(1,1,1), and an XGBoost model with lag features.

C. Validation Strategy

Classification models use an 80/20 train/test split and 5-fold stratified cross-validation, which preserves class proportions in every fold. We report accuracy, macro-averaged precision and recall, macro F1, and weighted F1. Regression uses MAE, RMSE, and R^2 . Forecasting uses a sequential split, training on January 2022 to September 2025 (45 months) and testing on October 2025 to September 2026 (12 months), and reports MAE, RMSE, and MAPE. Random shuffling is avoided for the series because it would leak future information into training. Because the largest class dominates postings, accuracy alone can mislead: the baseline’s 39% accuracy alongside a macro F1 of 0.08 illustrates this. We therefore treat macro F1 as the primary classification metric.

D. Implementation and Reproducibility

The system is implemented in Python 3.13 with scikit-learn, XGBoost, and Statsmodels, and the front end uses Streamlit 1.64.0 with Plotly charts. The code is organized into three modules (preprocessing, feature engineering, and models), seven notebooks that must be run in numerical order from data collection to error analysis, and a master script that executes cleaning, feature engineering, cross-validation, tuning, salary regression, forecasting, and figure generation in one command. The best classifier and the TF-IDF vectorizer are saved as artifacts so that the web application can serve predictions without retraining. Its output layer provides skill-demand rankings, salary distributions, a live job category predictor, and a salary estimator.

E. Error Analysis Procedure

Errors are analyzed in three ways: a confusion matrix of the tuned classifier on the test set, a review of misclassified test postings to identify recurring causes, and feature importances (Gini importance for Random Forest and gain for XGBoost).

IV. EXPERIMENTS AND RESULTS

A. Job Category Classification (RQ3)

Table IV compares baseline and advanced models. The majority-class baseline reaches 0.3911 test accuracy, but its macro F1 is only 0.0803 because it predicts one class out of seven; its macro recall of 0.1429 equals 1/7. All three tree-based models exceed 0.91 test accuracy. XGBoost is best on the test set (accuracy 0.9293, macro F1 0.9487), followed by Random Forest (0.9240, 0.9400). The tuned XGBoost attains the highest cross-validated accuracy (0.9355), although its cross-validated macro F1 (0.9355) is below that of untuned XGBoost (0.9416), and its test accuracy (0.9199) is lower than untuned XGBoost. Precision exceeds recall for all three tree-based models (for example 0.9554 versus 0.9425 for XGBoost), so on average a predicted category is slightly more reliable than it is exhaustive. XGBoost’s macro F1 (0.9487) is also above its weighted F1 (0.9298), which suggests that smaller classes are classified at least as well as large ones and that the model is not simply exploiting the largest class. Cross-validated and test accuracies

agree closely for the untuned models (0.9223 versus 0.9293 for XGBoost, 0.9168 versus 0.9240 for Random Forest), so there is no sign of overfitting to the training split. For the tuned model the order reverses: cross-validated accuracy exceeds test accuracy by 1.6 points, which would be consistent with mild optimism from selecting hyperparameters on the same folds. Fig. 4 shows the confusion matrix of the tuned model.

TABLE IV
MULTI-CLASS JOB CATEGORY CLASSIFICATION

Model	CV Acc.	CV F1	Test Acc.	Prec.	Rec.	Test F1	W-F1
Dummy	0.3912	0.0803	0.3911	0.0559	0.1429	0.0803	0.2199
Random Forest	0.9168	0.9304	0.9240	0.9477	0.9328	0.9400	0.9244
XGBoost	0.9223	0.9416	0.9293	0.9554	0.9425	0.9487	0.9298
Tuned XGBoost	0.9355	0.9355	0.9199	0.9496	0.9340	0.9413	0.9207

CV = 5-fold stratified cross-validation; Prec., Rec., and F1 are macro-averaged; W-F1 = weighted F1. Bold marks the best value in each column.

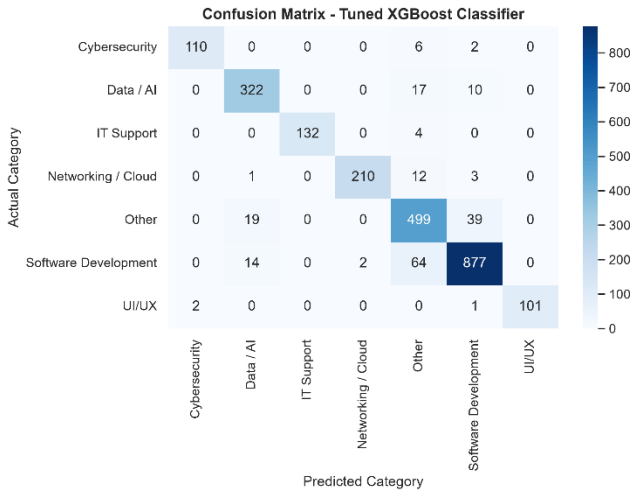


Fig. 4. Confusion matrix of the tuned XGBoost classifier over the job categories.

B. Salary Regression

Table V reports results on the 5,385 postings with disclosed salaries. Random Forest achieves an MAE of 7,944.77 BDT, an RMSE of 12,853.54 BDT, and $R^2 = 0.8422$, cutting MAE by about 68% relative to the median baseline (24,965.04 BDT). XGBoost is close (MAE 7,992.33 BDT, $R^2 = 0.8206$). Experience level, skill requirements, and role category therefore account for over 84% of salary variance in the best model (RQ2). For Random Forest, RMSE is about 1.6 times MAE, which indicates that a minority of larger misses inflates the squared error; XGBoost has a similar MAE but a higher RMSE (13,703.75 BDT), so its occasional errors are larger.

TABLE V
ADVERTISED MONTHLY SALARY REGRESSION (BDT)

Model	MAE	RMSE	R^2
Median baseline	24,965.04	32,410.31	-0.0036
Random Forest	7,944.77	12,853.54	0.8422
XGBoost	7,992.33	13,703.75	0.8206

Bold marks the best value in each column.

C. Demand Forecasting (RQ4)

Table VI reports forecasting results on the 12-month test period. The naive forecast has the lowest MAE (133.50) and MAPE (59.64%). ARIMA(1,1,1) has the lowest RMSE (222.91), which suggests it avoids the largest misses, yet its MAE (192.17) and MAPE (107.04%) are worse than the naive forecast. The XGBoost lag model is worst on all three metrics (MAE 334.50, RMSE 754.18, MAPE 194.67%). A naive last-value forecast needs no fitted parameters, whereas ARIMA and the XGBoost lag model must be estimated from only 45 points, which favors the simpler benchmark. Fig. 5 compares the forecasts with observed volumes.

TABLE VI
MONTHLY JOB DEMAND FORECASTING

Model	MAE	RMSE	MAPE (%)
Naive, $y_t = y_{t-1}$	133.50	278.79	59.64
ARIMA(1,1,1)	192.17	222.91	107.04
XGBoost lag model	334.50	754.18	194.67

MAE and RMSE are in monthly postings. Bold marks the best value in each column.

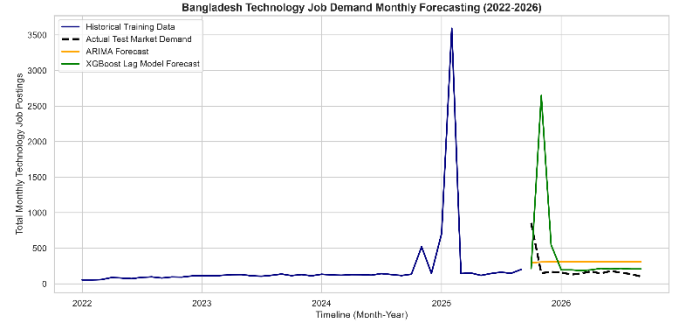


Fig. 5. Monthly job demand forecast comparison (2022–2026).

V. DISCUSSION AND ERROR ANALYSIS

A. What Worked

Skill indicators and TF-IDF terms separate technology roles well: XGBoost improves test accuracy by 53.8 percentage points over the majority-class baseline, and Random Forest is within 0.6 points of it (0.9240 versus 0.9293), so either ensemble is a practical choice. Salary is also predictable from the same kinds of features, with Random Forest cutting MAE by about 68% relative to the median baseline.

B. What Failed

Two results were negative. First, hyperparameter tuning raised cross-validated accuracy but not held-out accuracy, and the difference from untuned XGBoost is about one point, so we treat the two XGBoost variants as comparable rather than claiming a gain from tuning. Second, no forecasting model beat the naive baseline on MAE or MAPE. A likely reason is data volume: 45 monthly training points give a supervised model with lag features little to learn from, and the naive MAPE of nearly 60% indicates large month-to-month swings. Forecasts from this series should therefore be treated as exploratory.

C. Relation to Prior Work

We extend earlier descriptive and skill-trend analyses [1], [4] to 2026 data and to predictive tasks. Hoque and Islam found ARIMA models useful for tracking skill trends [1]; in our aggregate posting-volume series, a fixed-order ARIMA(1,1,1) improved RMSE but not MAE over the naive forecast, so its usefulness appears to depend on the target series and its length. Unlike the clustering approach of [4], we frame role identification as supervised classification, which allows direct evaluation against a baseline.

D. Error Analysis

Misclassified test instances trace to two main causes. The first is title and skill overlap: Data Analyst and Software Engineer postings sometimes share identical technical stacks, for example Python, SQL, and REST APIs, so skill features alone cannot separate them. The second is hybrid roles: Full Stack postings at smaller companies combine DevOps containerization with front-end development, which blurs the boundary between categories (Fig. 4).

E. Feature Importance

Fig. 6 ranks the top 15 features by Gini/gain importance. TF-IDF terms such as *development*, *cloud*, *ai*, *support*, and *cybersecurity* rank highly, alongside skill indicators for Figma, Python, and Docker. This pattern is plausible: category-specific vocabulary and tools identify categories, with Figma pointing to UI/UX work and Docker to cloud and DevOps work. Impurity-based importances can favor features with many split points and do not show the direction of an effect, so they support interpretation but do not prove it.

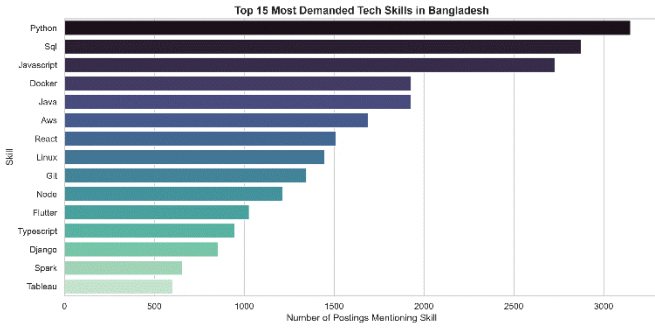


Fig. 6. Top 15 feature importances in job category classification.

F. Practical Implications

For students and job seekers, the ranking of Python, SQL, JavaScript, React, Docker, and AWS, together with salary ranges by experience level, gives a concrete basis for deciding what to learn. For universities and educators, the same evidence can support curriculum updates, for instance adding cloud and container tooling alongside programming fundamentals, as the demand for Docker (24.2%) and AWS (22.8%) suggests. For recruiters and HR leaders, the salary model and the classifier can benchmark packages in BDT and automate job description classification. For policymakers, the category shares and demand series offer an empirical complement to skills-gap reports [5], [6], although the forecasting results counsel caution in using the series for long-range planning.

G. Limitations

Six limitations qualify the results. (1) Category labels are derived from job-title rules rather than human annotation, so accuracy measures how well skill and text features recover this labeling scheme and may overstate performance against independently labeled data. (2) The missing-data defaults (experience, location, and posting date) simplify the data, and their sensitivity is not quantified. (3) Only about 44% of postings disclose a salary, the values are advertised rather than realized pay, and the disclosing subset may differ from the rest. (4) Deduplication on title, company, and location collapses roles reposted in different months, which may understate persistent demand in monthly counts. (5) The forecasting series has only 57 points. (6) Regex extraction covers 22 skills and can miss synonyms and context.

VI. CONCLUSION AND FUTURE WORK

TechCarrierBD shows that regex skill extraction and TF-IDF features with gradient-boosted trees classify technology postings into seven classes with 92.93% test accuracy and predict advertised monthly salary within a mean absolute error of 7,944.77 BDT. The framework unifies role classification, salary regression, and demand forecasting on 12,235 recent postings, and it reports failures openly: tuning did not improve the held-out score, and no forecasting model beat a naive baseline on MAE. For students and educators, the evidence points to Python, SQL, JavaScript, React, Docker, and AWS as the most requested skills.

Future work includes Transformer embeddings (BERT, RoBERTa) for fine-grained skill entity extraction and continuous real-time scraping across regional ICT hubs. We also plan human-annotated category labels to test the classifier independently of the title rules, longer series with seasonality-aware forecasting models, and a sensitivity analysis of the imputation defaults.

VII. AUTHOR CONTRIBUTIONS

Abdullah Al Fuwad contributed to data collection, preprocessing, and NLP skill-extraction and feature engineering. Md. Yeasin Arafat contributed to model development, evaluation and forecasting experiments, and the Streamlit application. Both authors contributed equally to the study design, analysis, manuscript writing, and final review.

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